Elizabeth (Liz) Liu

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Education

Columbia University New York, NY

Bachelor of Science in Operations Research, Minor in Artificial Intelligence & Psychology *Aug. 2025 – May 2029*

- GPA: 3.6871/4.00
- Relevant Coursework: Data Structures and Algorithms, Multivariable Calculus, Data Engineering, Probability
- Activities: Society of Women Engineers (Secretary), Jumpstarting Aspiring Developers and Entrepreneurs, Institute for Operations Research and the Management Sciences (Treasurer), Greenborough Special Interest Community (Co-President)

Experience

Head Intern and Economic Research Analyst Intern Jun. 2023 – Aug. 2025

City of Beachwood Economic Development Department *Beachwood, OH*

- Mentored and supervised a team of 12 interns as the organization's most senior intern
- Led research and planning for a biomedical innovation ecosystem that directly supported growth of industry clusters, generating 100+ jobs, $320K+ in City tax benefits, and $5M in capital investment over 3 years
- Laid the foundation for the Beachwood Community Improvement Corporation's innovation committee, which has now funded $60K in forgivable loans for 3 R&D companies and launched a $1M revolving loan program

Marketing Intern Feb. 2023 – Jan. 2025

JNForensics, LLC *Tampa, FL (Remote)*

- Produced marketing content featured by Joe Navarro on Twitter, LinkedIn, Facebook (73.1K+ followers)
- Enhanced understanding of nonverbal communication, branding, and audience psychology through mentorship from Joe Navarro (CEO, JNForensics; former FBI Special Agent) and Anne-Maartje Oud (CEO, Behaviour Company)

AI and Machine Learning Intern May 2024 – Aug. 2024

Case Western Reserve University School of Engineering *Cleveland, OH*

- Developed Python models using logistic regression, Gaussian processes, and Bayesian optimization to analyze uncertainty and support predictive decision-making

Cambridge Future Scholar Programme Jun. 2023 – Aug. 2023

Cambridge Centre for International Research | Paper *Cambridge, UK (Remote)*

- Synthesized competing historical variables and strategic frameworks to construct a structured analytical argument
- Applied game-theory concepts (signaling, commitment problems, equilibrium failures) to analyze complex decision-making dynamics; worked with Dr. Stephen Chaudoin (Harvard) and Dr. Michael-David Mangini (Yale)

Leadership & Activities

Columbia Society of Women Engineers Sep. 2025 – Present

Secretary *Apr. 2026 – Present*

- Track and analyze member engagement data to inform programming decisions of 50 person leadership team
- Author the weekly newsletter, serving as the organization's primary communications channel to 1.4K+ members

Organizational Committee Member *Sep. 2025 – Apr. 2026*

- Developing Engineering Exploration Experience event, hosting 200+ high schoolers; part of a 6-member team
- Leading outreach and recruitment efforts; secured 3 high-profile speakers (keynote and 2 industry panelists)

Columbia Space Initiative | *NASA SUITS Project* | *Navigation Team Lead* Sep. 2025 – Dec. 2025

- Secured acceptance of Columbia's NASA SUITS proposal by aligning 6 subteams around a cohesive submission

Hope Heroes Ohio | *VP of Finance* Aug. 2024 – Jul. 2025

- Managed finances and pitch strategy for a 41-member team, coordinating fundraising that generated $238K+ in corporate sponsorships for children's hospitals in Northeast Ohio

Skills, Honors & Interests

Technical Skills: Python (Proficient), Java (Proficient), C++ (Intermediate), SQL (Intermediate), Stata (Intermediate), NumPy, Matplotlib, pandas, Microsoft Office, Google Workspace, Salesforce CRM
Languages: English (native), Mandarin Chinese (conversational, heritage), Spanish (formerly proficient, inactive)
Interests: classical violin (13 years), film analysis, gastronomy, human behavior, sculptural ceramics, strength training